Power Rule

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\int x^n\,dx = \begin{cases} \frac{x^{n+1}}{n+1} + C, & n \neq -1, \\ \ln|x| + C, & n = -1 \end{cases}$$

Scalar Multiple Rule

$$\frac{d}{dx}[c\,f(x)] = c\,f'(x)$$

$$\int c\,f(x)\,dx = c\int f(x)\,dx$$

Sum and Difference Rule

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

$$\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

Product Rule

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

No general integral rule exists. Integration by parts is related but not a strict inverse.

Quotient Rule

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

No general integral rule exists.

Chain Rule

$$\frac{d}{dx}[f(g(x))] = f'(g(x))\,g'(x)$$

No single inverse rule. (Substitution is the corresponding integral technique.)

Common Functions

$$\frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(a^x) = a^x \ln a,$$

$$\int e^x dx = e^x + C, \quad \int a^x dx = \frac{a^x}{\ln a} + C$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}, \quad \frac{d}{dx}(\sin x) = \cos x,$$

$$\int \frac{1}{x} dx = \ln|x| + C, \quad \int \cos x\,dx = \sin x + C$$

$$\frac{d}{dx}(\cos x) = -\sin x, \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

$$\int -\sin x\,dx = \cos x + C, \quad \int \sec^2 x\,dx = \tan x + C$$